

EXPERIMENT: 7

ANALOG BEAMFORMING ANALYSIS USING MATLAB

Objective – 1: Analyze Beamforming Gain (dB)

MATLAB Program

```
clc;

clear;

close all;

antennas = [2 4 8 16 32]; % Number of Antenna Elements

gain = 10*log10(antennas); % Beamforming Gain (dB)

figure

bar(antennas, gain)

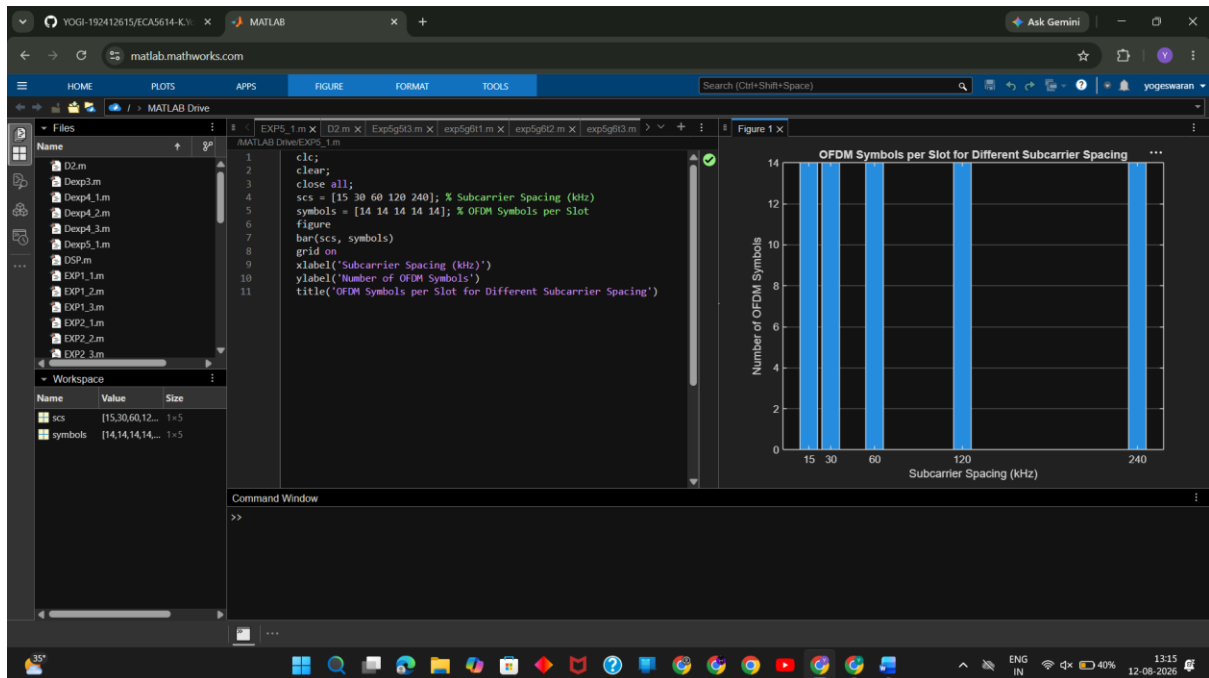
grid on

xlabel('Number of Antenna Elements')

ylabel('Beamforming Gain (dB)')

title('Beamforming Gain for Different Antenna Array Sizes')
```

OUTPUT



Objective – 2 Compare Beam Width (Degrees)

Matlab Code :

```
clc;

clear;

close all;

antennas = [2 4 8 16 32]; % Number of Antenna Elements

beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)

figure

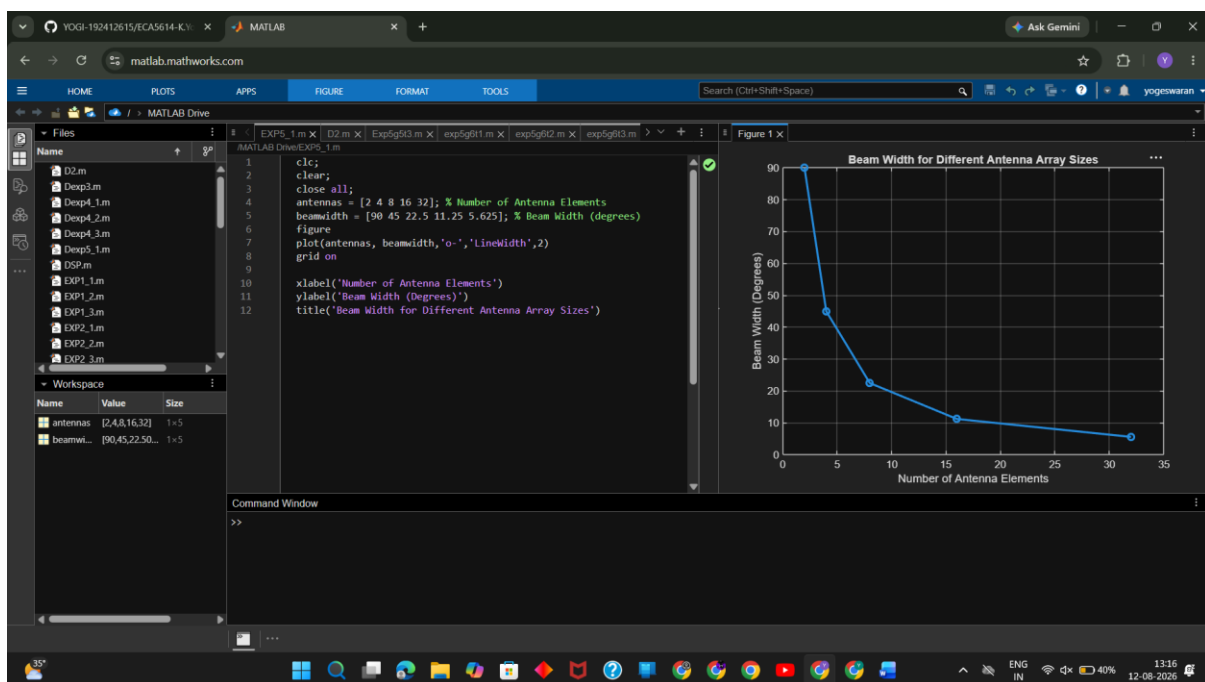
plot(antennas, beamwidth,'o-','LineWidth',2)

grid on

xlabel('Number of Antenna Elements')

ylabel('Beam Width (Degrees)')

title('Beam Width for Different Antenna Array Sizes')OUTPUT
```



Objective – 3 Analyze Main Lobe Direction (Degrees)

Matlab Code

```
clc;

clear;

close all;

theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)
```

```
gain = [1 1 1 1 1]; % Constant normalized gain
```

figure

```
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
```

```
title('Main Lobe Direction in Analog Beamforming')
```

OUTPUT

